SG No. 188 D_{3h}^2 $P\bar{6}c2$ [hexagonal]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `2a`, site symmetry: `3.2`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `3.2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,3,-10,-11,-12]`
2	$[0, 0, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4,5,6,-7,-8,-9]`

Table 2: Wyckoff bond: `6b@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, 0]$	`[1,-11]`
2	$[0, X, Z]$	$[0, 0, 0]$	`[2,-10]`
3	$[-X, -X, Z]$	$[0, 0, 0]$	`[3,-12]`
4	$[X, 0, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4,-8]`
5	$[0, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[5,-7]`
6	$[-X, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[6,-9]`

Table 3: Wyckoff bond: `6c@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[0, 0, 0]$	`[1,11]`
2	$[-2X, -X, 0]$	$[0, 0, 0]$	`[2,10]`
3	$[X, -X, 0]$	$[0, 0, 0]$	`[3,12]`
4	$[X, 2X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[4,8]`
5	$[-2X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[5,7]`
6	$[X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[6,9]`

Table 4: Wyckoff bond: `12d@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, 0]$	`[3]`
4	$[X, Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[7]`
8	$[-X+Y, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[8]`
9	$[X, X-Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, 0, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[0, 0, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[0, 0, 0]$	`[12]`

* Wyckoff site: `2b`, site symmetry: `-6..`

Table 5: Wyckoff bond: `2a@2b`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,-4,-5,-6]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7,8,9,-10,-11,-12]`

Table 6: Wyckoff bond: `6b@2b`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2,5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3,6]`
4	$[-Y,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7,10]`
5	$[-X+Y,\ Y,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[8,11]`
6	$[X,\ X-Y,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,12]`

Table 7: Wyckoff bond: `12c@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3]`
4	$[X,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[4]`
5	$[-Y,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[5]`
6	$[-X+Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7]`
8	$[-X+Y,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[8]`
9	$[X,\ X-Y,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[12]`

* Wyckoff site: `2c`, site symmetry: `3.2`

Table 8: Wyckoff bond: `2a@2c`, bond symmetry: `3.2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[4,5,6,-7,-8,-9]`

Table 9: Wyckoff bond: `6b@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[3,-12]`

continued ...

Table 9

No.	vector	center	mapping
4	$[X, 0, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[4,-8]`
5	$[0, X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[5,-7]`
6	$[-X, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[6,-9]`

Table 10: Wyckoff bond: `6c@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,11]`
2	$[-2X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[2,10]`
3	$[X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[3,12]`
4	$[X, 2X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[4,8]`
5	$[-2X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[5,7]`
6	$[X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[6,9]`

Table 11: Wyckoff bond: `12d@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[3]`
4	$[X, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[7]`
8	$[-X+Y, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[8]`
9	$[X, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[12]`

* Wyckoff site: `2d`, site symmetry: `-6..`

Table 12: Wyckoff bond: `2a@2d`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,2,3,-4,-5,-6]`
2	$[0, 0, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[7,8,9,-10,-11,-12]`

Table 13: Wyckoff bond: `6b@2d`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,4]`
2	$[-Y, X-Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[2,5]`

continued ...

Table 13

No.	vector	center	mapping
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[3,6]
4	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[7,10]
5	$[-X+Y,\ Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[8,11]
6	$[X,\ X-Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[9,12]

Table 14: Wyckoff bond: 12c@2d, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[1]
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[3]
4	$[X,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[4]
5	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[5]
6	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[6]
7	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[7]
8	$[-X+Y,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[8]
9	$[X,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[9]
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[10]
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[11]
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[12]

* Wyckoff site: 2e, site symmetry: 3.2

Table 15: Wyckoff bond: 2a@2e, bond symmetry: 3.2

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[1,2,3,-10,-11,-12]
2	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[4,5,6,-7,-8,-9]

Table 16: Wyckoff bond: 6b@2e, bond symmetry: ..2

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[1,-11]
2	$[0,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[2,-10]
3	$[-X,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[3,-12]
4	$[X,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[4,-8]
5	$[0,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[5,-7]
6	$[-X,\ -X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[6,-9]

Table 17: Wyckoff bond: 6c@2e, bond symmetry: ..2

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[1,11]

continued ...

Table 17

No.	vector	center	mapping
2	$[-2X, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[2,10]`
3	$[X, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[3,12]`
4	$[X, 2X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[4,8]`
5	$[-2X, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[5,7]`
6	$[X, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[6,9]`

Table 18: Wyckoff bond: `12d@2e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[3]`
4	$[X, Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[7]`
8	$[-X+Y, Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[8]`
9	$[X, X-Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[12]`

* Wyckoff site: `2f`, site symmetry: `-6..`

Table 19: Wyckoff bond: `2a@2f`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[1,2,3,-4,-5,-6]`
2	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[7,8,9,-10,-11,-12]`

Table 20: Wyckoff bond: `6b@2f`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[1,4]`
2	$[-Y, X-Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[2,5]`
3	$[-X+Y, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[3,6]`
4	$[-Y, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[7,10]`
5	$[-X+Y, Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[8,11]`
6	$[X, X-Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[9,12]`

Table 21: Wyckoff bond: `12c@2f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[3]`
4	$[X, Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[7]`
8	$[-X+Y, Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[8]`
9	$[X, X-Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[12]`

* Wyckoff site: `4g`, site symmetry: `3..`

Table 22: Wyckoff bond: `4a@4g`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2,3]`
2	$[0, 0, -Z]$	$\left[0, 0, \frac{1}{2}-z\right]$	`[4,5,6]`
3	$[0, 0, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[7,8,9]`
4	$[0, 0, -Z]$	$[0, 0, -z]$	`[10,11,12]`

Table 23: Wyckoff bond: `12b@4g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, z]$	`[3]`
4	$[X, Y, -Z]$	$\left[0, 0, \frac{1}{2}-z\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[0, 0, \frac{1}{2}-z\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}-z\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[7]`
8	$[-X+Y, Y, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[8]`
9	$[X, X-Y, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, 0, -z]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[0, 0, -z]$	`[11]`
12	$[X, X-Y, -Z]$	$[0, 0, -z]$	`[12]`

* Wyckoff site: `4h`, site symmetry: `3..`

Table 24: Wyckoff bond: `4a@4h`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3]`

continued ...

Table 24

No.	vector	center	mapping
2	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[4,5,6]`
3	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[7,8,9]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[10,11,12]`

Table 25: Wyckoff bond: `12b@4h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3]`
4	$[X,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[4]`
5	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[5]`
6	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[7]`
8	$[-X+Y,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[8]`
9	$[X,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[12]`

* Wyckoff site: `4i`, site symmetry: `3..`

Table 26: Wyckoff bond: `4a@4i`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[1,2,3]`
2	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[4,5,6]`
3	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[7,8,9]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[10,11,12]`

Table 27: Wyckoff bond: `12b@4i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[3]`
4	$[X,\ Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[4]`
5	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[5]`
6	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[7]`
8	$[-X+Y,\ Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[8]`
9	$[X,\ X-Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[11]`

continued ...

Table 27

No.	vector	center	mapping
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[12]`

* Wyckoff site: `6j`, site symmetry: `..2`

Table 28: Wyckoff bond: `6a@6j`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ Z]$	$[x,\ -x,\ 0]$	`[1,-10]`
2	$[-X,\ 0,\ Z]$	$[x,\ 2x,\ 0]$	`[2,-12]`
3	$[0,\ -X,\ Z]$	$[-2x,\ -x,\ 0]$	`[3,-11]`
4	$[X,\ X,\ -Z]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[4,-7]`
5	$[-X,\ 0,\ -Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[5,-9]`
6	$[0,\ -X,\ -Z]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[6,-8]`

Table 29: Wyckoff bond: `6b@6j`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ -X,\ 0]$	$[x,\ -x,\ 0]$	`[1,10]`
2	$[X,\ 2X,\ 0]$	$[x,\ 2x,\ 0]$	`[2,12]`
3	$[-2X,\ -X,\ 0]$	$[-2x,\ -x,\ 0]$	`[3,11]`
4	$[X,\ -X,\ 0]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[4,7]`
5	$[X,\ 2X,\ 0]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[5,9]`
6	$[-2X,\ -X,\ 0]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[6,8]`

Table 30: Wyckoff bond: `12c@6j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ -x,\ 0]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[x,\ 2x,\ 0]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[-2x,\ -x,\ 0]$	`[3]`
4	$[X,\ Y,\ -Z]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[4]`
5	$[-Y,\ X-Y,\ -Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[5]`
6	$[-X+Y,\ -X,\ -Z]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[7]`
8	$[-X+Y,\ Y,\ Z]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[8]`
9	$[X,\ X-Y,\ Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$[x,\ -x,\ 0]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$[-2x,\ -x,\ 0]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$[x,\ 2x,\ 0]$	`[12]`

* Wyckoff site: `6k`, site symmetry: `m..`

Table 31: Wyckoff bond: `6a@6k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, y, \frac{1}{4}\right]$	`[1,4]`
2	$[-Y, X-Y, 0]$	$\left[-y, x-y, \frac{1}{4}\right]$	`[2,5]`
3	$[-X+Y, -X, 0]$	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3,6]`
4	$[-Y, -X, 0]$	$\left[-y, -x, \frac{3}{4}\right]$	`[7,10]`
5	$[-X+Y, Y, 0]$	$\left[-x+y, y, \frac{3}{4}\right]$	`[8,11]`
6	$[X, X-Y, 0]$	$\left[x, x-y, \frac{3}{4}\right]$	`[9,12]`

Table 32: Wyckoff bond: `6b@6k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, y, \frac{1}{4}\right]$	`[1,-4]`
2	$[0, 0, Z]$	$\left[-y, x-y, \frac{1}{4}\right]$	`[2,-5]`
3	$[0, 0, Z]$	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3,-6]`
4	$[0, 0, Z]$	$\left[-y, -x, \frac{3}{4}\right]$	`[7,-10]`
5	$[0, 0, Z]$	$\left[-x+y, y, \frac{3}{4}\right]$	`[8,-11]`
6	$[0, 0, Z]$	$\left[x, x-y, \frac{3}{4}\right]$	`[9,-12]`

Table 33: Wyckoff bond: `12c@6k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, y, \frac{1}{4}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[-y, x-y, \frac{1}{4}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3]`
4	$[X, Y, -Z]$	$\left[x, y, \frac{1}{4}\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[-y, x-y, \frac{1}{4}\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[-x+y, -x, \frac{1}{4}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[-y, -x, \frac{3}{4}\right]$	`[7]`
8	$[-X+Y, Y, Z]$	$\left[-x+y, y, \frac{3}{4}\right]$	`[8]`
9	$[X, X-Y, Z]$	$\left[x, x-y, \frac{3}{4}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[-y, -x, \frac{3}{4}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[-x+y, y, \frac{3}{4}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[x, x-y, \frac{3}{4}\right]$	`[12]`

* Wyckoff site: `12l`, site symmetry: `1`

Table 34: Wyckoff bond: `12a@12l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-y, x-y, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x+y, -x, z]$	`[3]`
4	$[X, Y, -Z]$	$\left[x, y, \frac{1}{2}-z\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[-y, x-y, \frac{1}{2}-z\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[-x+y, -x, \frac{1}{2}-z\right]$	`[6]`

continued ...

Table 34

No.	vector	center	mapping
7	$[-Y,\, -X,\, Z]$	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[7]
8	$[-X+Y,\, Y,\, Z]$	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	[8]
9	$[X,\, X-Y,\, Z]$	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	[9]
10	$[-Y,\, -X,\, -Z]$	$[-y,\, -x,\, -z]$	[10]
11	$[-X+Y,\, Y,\, -Z]$	$[-x+y,\, y,\, -z]$	[11]
12	$[X,\, X-Y,\, -Z]$	$[x,\, x-y,\, -z]$	[12]